# Rise & Run

a card game about straight lines · 2–6 players

## The aim

Be the first player to play all of your cards. Each card shows a straight line, with its equation and graph. Match the line in the middle in the way the die tells you.

## You need

The Rise & Run deck (44 cards) and one standard die.

## Setting up

Shuffle the cards and deal 5 to each player. Turn one card face up in the middle to start the discard pile, and put the rest face down as the draw pile. The youngest player starts, and play passes clockwise.

## Taking a turn

Roll the die and see what you get. On a 2, 3, 4 or 5, the number tells you how your card has to match the top card of the discard pile.

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| **1** | **Draw!** | Take one card from the draw pile. Your turn is over. |
| **2** | **Parallel.** | Your line is parallel to the top card (they never meet). |
| **3** | **Perpendicular.** | Your line crosses the top card at a right angle. |
| **4** | **Same y-intercept.** | Both lines cross the y-axis at the same point. |
| **5** | **Same x-intercept.** | Both lines cross the x-axis at the same point. |
| **6** | **Discard!** | Choose any card from your hand and put it on the discard pile. It becomes the new top card. |

- **Rolled 2–5 and got a match?** Play it face up on the discard pile. It becomes the new top card. Say why it matches, for example “Both have gradient 2”. If you have more than one match, you choose.
- **No match?** Your turn is simply over. You don’t draw a card.
- **Challenge!** If a player thinks a card doesn’t match, they can say so. Check the graphs together. If the card doesn’t match, its player takes it back and draws a card. If it does match, the challenger draws a card. (A card discarded on a 6 never needs a match.)
- **Running out?** If the draw pile is empty, shuffle the discard pile, keeping the top card in the middle, and carry on.

## Special lines

- Horizontal lines (like $y = 3$) are parallel to each other. Vertical lines (like $x = 3$) are parallel to each other. A horizontal line and a vertical line are perpendicular.
- A horizontal line never crosses the x-axis, and a vertical line never crosses the y-axis. They have no intercept there, so they can’t be matched on that intercept.
- **$y = 0$** is a wildcard for roll 5. It lies along the x-axis, so it matches any x-intercept: play it on any top card that has an x-intercept, or play any such card on top of it. The only exception is another horizontal line (like $y = 3$), which never crosses the x-axis.
- **$x = 0$** is a wildcard for roll 4. It lies along the y-axis, so it matches any y-intercept, in the same way. The only exception is a different vertical line (like $x = 3$), which never crosses the y-axis.

## Winning

The first player to get rid of their last card, by playing it or by discarding it, wins! A game usually takes about 10 minutes.

## Maths hints

- The gradient ($m$ in $y = mx + c$) tells you how steep a line is. Lines with the same gradient are parallel.
- Perpendicular lines have gradients that multiply to $-1$. A line with gradient 2 is perpendicular to one with gradient $-0.5$, and gradient 1 is perpendicular to gradient $-1$.
- The y-intercept is the $c$ in $y = mx + c$. For the x-intercept, put $y = 0$ and solve, or read it off the graph.

**Easy mode:** play with the Easy Mode deck, where every card shows its intercepts on the graph and underneath it (**blue** for the y-intercept, **orange** for the x-intercept). **Challenge mode:** once you’re confident, switch to the standard deck, where you have to read the intercepts off the graph yourself.